Course Introduction

EXPECTATIONS, FAILURE, FUN

Who Am I?

Jack Forden

- Marquette alumni ‘22, -> Graduate Program: ‘25
- Focus Areas:
  - CS-Education, Student Retention, Languages, CS History, OS, Compilers
- Outside of Academics
  - Competitive Games +++
    - CS2
    - Tarkov
  - Chillies aficionado
  - Favorite things:
    - FMAB
    - Wilder Woods(FEVER/Sky, Patience)
  - Toes in the water
    - Game Design
    - Tournament organizing for Broken Arrow
      - https://www.ba-hub.net/
    - Extreme couponing

What is this course?

What Does Success in This Course Mean?

- Success in this course is **not** about making money
  - That would be nice though…
- Success in this course is about **learning how systems fail**
- This course's success metrics:
  - We build something that executes
  - We understand why we failed in the past
  - We learn the methodology
    - Data cleaning and feature engineering
    - Signal processing concepts
    - System design thinking
    - Risk management basics
- You leave this course knowing **exactly** why get rich quick schemes don't work
  - Well…because we are going to build one

Why Trading is Hard

- Everything works against you
- Speed
  - HFT firms spend millions to save milliseconds
  - Our systems are going to take seconds or longer
- Information Asymmetry
  - The price we see is a memory, not reality
  - Order books constantly shift and move
- The Noise Problem
  - Our brain wants to see patterns that don’t exist
  - Random walks look like trends
- Psychology
  - We will watch our money evaporate in real-time
  - Every red bar feels personal

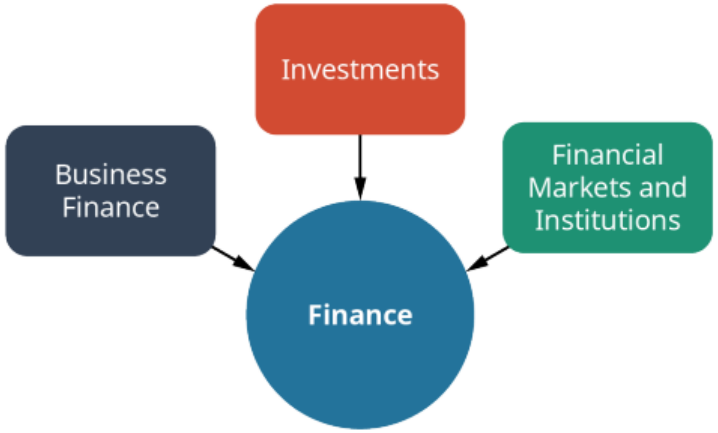
Why Automation Fails Faster

- Automation doesn’t fix problems – it amplifies them
- Human Trading Failures
  - Emotional Decisions
  - Second Guessing
  - Fatigue
- Automated Trading Failure Modes
  - Bad Logic
  - No emotional circuit breaker
    - Ride losses to zero
  - Perfect Consistency
    - Wrong at high speeds

What is Finance in this course?

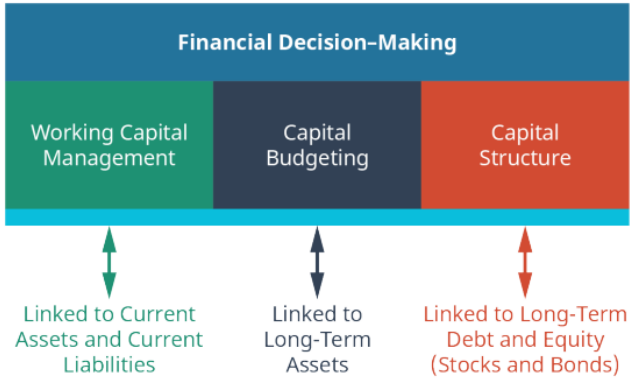
What is Finance as a whole?

- Finance
  - The study of the management, movement, and raising of money
  - The study of understanding the uses and sources of cash, as well as the concept of a risk-reward trade-off
  - Finance is a tool that can help us be better decision makers
- Finance is divided into three primary areas in the domestic market
  - Business finance
  - Investments
  - Financial markets and institutions



What is Finance: Business Finance

- **Business Finance** looks at how managers can apply financial principles to maximize the value of a firm in a risky environment
  - In the case of corporations, shareholders own the company who hire managers to run the company with the intent to maximize shareholder wealth
    - *How does this decision impact the wealth of the shareholders*
- In business finance, managers focus on three broad areas
  - **1. Working Capital Management** is the study and management of short-term assets and liabilities
    - The CFO and finance team set company policy on how to manage WCM
    - Determines credit policy, establishes minimum criteria for extension of credit, terms of lending, etc.



What is Finance: Business Finance

- 2. **Capital budgeting** is the process of determining which long-term or fixed assets to acquire in an effort to maximize shareholder value.
  - Decisions ideally add the greatest value to a firm
  - Estimating the value of potential investments by forecasting the size, timing, and risk of cash flows associated with the investments
  - Finance departments develop and compile cash flow estimates to develop a portfolio of investment projects that maximize the value of the firm
- 3. **Capital structure** is the process by which managers focus more specifically on long-term debt and increasing shareholder wealth.
  - Capital structure questions require financial managers to work with economists, lenders, underwriters, investment bankers, and other sources of external financial information and financial capital

What is Finance: Investments

- **Investments** are products and processes used to create and grow wealth
  - Application of the different types of financial instruments, delivery vehicles, regulation, and risk-and-return opportunities
  - Stocks, bonds, derivative securities(futures and options)
- Investments also include mutual funds, exchange-traded funds(ETFs), and investment vehicles such as
  - 401k, 403b, Roth, IRA's
- Real Assets such as Gold, real estate, and commodities are also common investment opportunities

What is Finance: Financial Markets and Institutions

- **Financial Markets and Institutions** are the firms and regulatory agencies that oversee our financial system
  - There is *significant* overlap with the prior two categories of finance
    - A firm that is involved in profit seeking ultimately requires good financial management
- *Much* of the US regulatory structure for financial markets and institutions developed in the 1930's as a response to the stock market crash of 1929 and the subsequent great depression.
- In the US, the desire for safety and protection of investors and the financial industry led to the development of
  - Security and Exchange Commission
    - Formed from the Securities Act of 1933 and 1934
  - Major bank regulation in 1933 and 1935 gave rise to government backed bank deposit insurance(FDIC)
- These regulatory acts separated investment banking from commercial banking.

Why Study Finance?

- Finance is the lubricant that keeps our economy running smoothly
- There are **many** career opportunities in the field of finance
  - BLS predicts that close to an additional 500,000 new finance and accounting related jobs will be created by 2029
- **Traditional Financial Roles:**
  - Financial Manager, Investment relations associate, budget analysist, credit analyst, finance analyst, personal financial advisor, loan officer, insurance underwriter, financial examiner, finance professor
- **Immerging Financial Roles**
  - Digital Transformation Analyst, regulatory technology specialist, Cryptocurrency & Digital Asset Strategist, AI-Driven Financial Risk Modeler, FinTech Product Manager, Blockchain & Smart-Contract Auditor, Real-Time Financial Data Analyst

Why Study Finance?

- A career in finance is just **one** reason to study finance
- Finance is an *excellent* decision making tool
  - Analytical thinking
  - Life and asset planning
- An understanding of finance and financial markets opens a broader world of available financial investment opportunities
  - It also helps us plan our lives
  - understand how to make smarter decisions

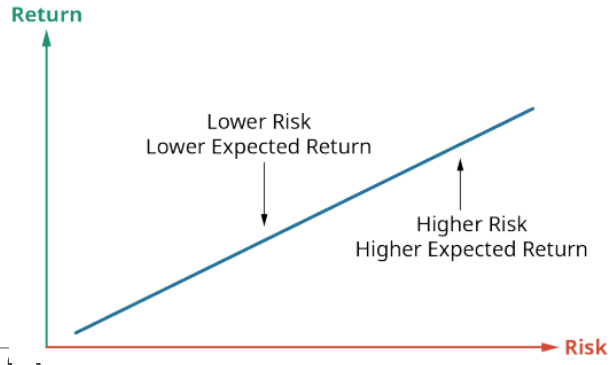
Finance as Coordination Under Uncertainty

FINANCE IS HOW SOCIETIES COORDINATE DECISIONS ABOUT SCARCE RESOURCES

Risk in Finance

- Finance tells us that an increase in risk results in an increase in expected return
  - **Risk** at its most basic level is **uncertainty**
- The study of finance attempts to *quantify* risk in a way that helps individuals and organizations assess an appropriate trade-off for risk
- Risk-Return tradeoffs are all around us in our everyday decision-making
  - Jaywalk or walk down to the crosswalk
  - Accept the first job offer after graduation or hold out for the offer you *really* want?
- A better understanding of finance makes these types of decisions easier

Return in Finance



- **Return** is compensation for making an investment and waiting for the benefit
  - Interest earned on an investment in a bond
  - Dividend from the purchase of a stock
  - Higher income received and greater job satisfaction from a degree
- Investors would not be satisfied if the average return on stocks and bonds were the same as that for a risk-free savings account
  - Stocks and bonds have a higher risk, thus investors expect a greater average return
- The study of finance provides us with the tools to make better and more consistent assessments of the risk-return tradeoff in all decision making
  - Portfolios of investment securities tend to demonstrate the characteristics of a normal return distribution
- There are **many** types of risk: default, inflation, diversifiable, political
  - The study of finance helps us better navigate and plan

Risk vs Return

- This semester is **fraught** with *risk*
- The technologies we are going to engage with are *experimental* and often *untested* for the type of work we are trying to do
  - No textbooks
  - No resources
  - All exploration
- The **Risk**
  - Demo's don't work as we expect
  - The technology isn't as robust or able to produce the type of results we may hope
  - It doesn't work :/
- The **Reward**
  - We get super rich and it works super well (not going to happen)
  - We learn about our rapidly evolving digital landscape(Agentic Systems)
  - We learn fundamentals about trading (basics and quantitative methods)
  - We learn the methodologies (data cleaning, financial analysis, statistical methods)

Who is in the Market

WHAT DO WE EVEN MEAN WHEN WE TALK ABOUT THE “MARKET”

The “Market”

- Our *environment* for this semester is **complex**
- The market is essentially the **intersection** where buyers and sellers congregate to trade assets
  - A market can be a store, Facebook *marketplace*, or in our case **security exchanges**
- This course is going to view the market as a high-dimensional, real-time data environment
  - Every “tick” will represent a decision made
    - By a human, agent, or hybrid
- The simplistic view of our environment is purely the security exchanges(NYSE, NASDAQ)
  - This view undercuts the *players* that drive the markets
    - They players have a direct impact on the environment we are trying to understand
- Players
  - Institutional/market makers - **Dealers**
    - Provides the pipes, execute trades, report data
  - **Brokers**
    - Those that enable trades in markets to happen, match buyers and sellers
  - Speculators(us)
    - Retail investors
    - Quants who seek returns through pattern recognition
    - Students who want to learn

The “Market” and it’s many Participants

- Primary market is for *new* securities
  - New shares are issues when a firm first enters the stock market via IPO(initial public offering)
    - Once the initial purchasing happens, these purchasers may then trade them
- Secondary market is for *used* securities
- The stock market largely deals with <u>used</u> securities
  - Stock markets trade equity securities daily
    - Most of this takes place among individual and institutional investors who own shares in publicly traded companies
    - These trades have little effect on the firm itself; however, the information provide by transactions is costly
      - Real time statement by investors of the firm's perceptions
  - Companies are responsible for monitoring and publishing **billions** of trades across multiple markets
    - Every second hundreds of thousands of trades are executed

The Major Players: NYSE and NASDAQ

- NYSE(New York Stock Exchange)
  - Founded in 1792 the NYSE is the worlds largest stock exchange
  - Designated Market Makers act as auctioneers to ensure fair trade
  - Hybrid approach
    - NYSE is one of the few remaining trading floors where humans are still actively trading in person
  - Traditionally home to "Blue Chip" industrial giants like Coca-Cola or Ford.
- NASDAQ(National Association of Securities Dealers)
  - Founded in 1971 as the first fully electronic stock market
  - No physical trading floor
  - Known for high-growth tech companies like Apple, Microsoft, and Google
- Both of these markets allow companies to go public and raise money for growth by selling shares to investors
  - Who then can trade the shares
  - They NYSE and NASDAQ are often reflectors of the overall health and sentiment of the global economy

Key Market Players

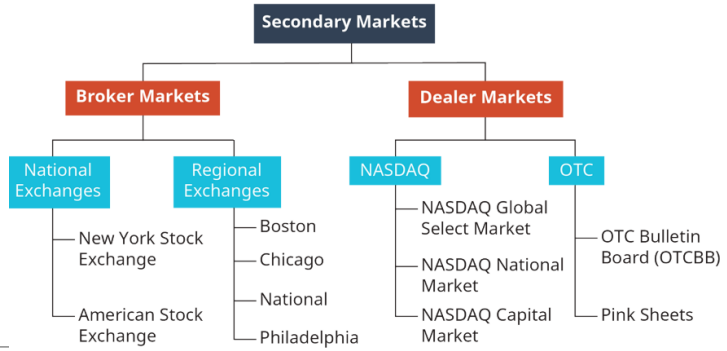


Figure 1.7 Broker and Dealer Markets

- **Dealers**
  - Own the securities that they buy or sell. When a dealer engages in a financial transaction, they are trading from their own portfolio.
  - Dealers do *not* participate in the market in the same manner as an individual or institutional investor
    - Dealers do not try to make money by ensuring their investments are worth as much as possible. Instead, dealers attempt to "make markets"
      - They are willing and able to buy and sell at the current bid and ask price
        - Dealers make money from the volume of trading and the spread between their bid price(what they are willing to pay) and their ask price(price they are willing to sell)
- Let's look at how this works in practice

Market Maker – Citadel Securities

- When you *sell* a stock, you are typically not selling to another retail investor
  - If you use a brokerage app to sell a share of Apple stock, you usually aren't selling it directly to another person. You are selling it to a dealer like Citadel.
    - Citadel buys at your asking price, then immediately tries to sell it to someone else at a slightly higher price to capture the *spread*
      - **Spread:** the movement of a stocks price from *buy* to *sell* price – will go over in more detail in our next lecture
- **Citadel**
  - **#1** in retail market making with ~35% of US listed retail volume through the platform
  - **645 billion** in trades executed daily
    - Citadel might offer to buy a stock at $100.00 and they might sell it at $100.01
      - When you execute billions of trades a day, it adds up

Brokers

- Brokers act as facilitators in a market
  - They bring together buyers and sellers for a transaction
- Brokers typically act on commission from a sale of an asset
- Brokers are required for clients because security exchanges require membership in the exchange to accept orders
  - A seat on an exchange last sold for 3.7 million
    - Seats have been absolved to now revolve around monthly membership fees at about $7,200 per year
- Discount brokers are increasingly more popular
  - Discount brokers execute trades for clients while offering very low or no commission on many of their trade executions
    - Robin hood
      - *Gives* the trade to citadel(who pays them)
- How does this work in practice?

Dealers and Brokers

- Steps and how each company makes money
  - 1. I want to sell 1 share of a company for $50.00, this is the **bid price**
  - 2. I click sell !
  - 3. Robinhood's algorithm sees that Citadel Securities is offering the best deal right now, so we ping citadels servers
  - 4. By law, Citadel must give you the "Best Execution." If the public exchange (NYSE) is buying at $50.00, Citadel might actually buy it from you for $50.01
    - Why? To keep Robinhood happy and keep the regulators off their back, they often give you a slightly better price than the "official" one. You just made an extra penny.
  - 5. Citadel now owns your share for $50.01. They immediately (in microseconds) try to sell it to a buyer for **$50.02**.
    - They keep the **$0.01 spread** as their "fee" for taking the risk of holding that share.
  - 6. Citadel then sends a tiny fraction of a cent (e.g., $0.001) back to Robinhood as a "thank you" for sending them that trade.
- This happens for every trade, that's every transaction, billions of times a day

Financial Intermediaries

- Commercial banks or mutual fund investment companies
  - Act as an intermediary to enable easier and more efficient exchanges among transacting parties
- A commercial bank accepts deposits from savers and investors and creates loans for borrowers
- An investment company pools funds from investors to inexpensively purchase and manage portfolios of stock and bonds
  - This is noticeably different from brokers and dealers selling their own portfolios
  - Financial intermediaries may accept money from investors and may create a completely different security all together
    - We will discuss mutual funds in greater depth next week
- Large institutions can spread fixed costs (like technology, regulatory compliance, and staff salaries) over millions of customers, making services cheaper for everyone.

Market Players, Managers, and you

- Changes in the *environment* are felt by all players in a market
  - Brokers, Dealers, Financial Intermediaries, and retail investors
- Throughout this semester we are going to try to **quantify** the *scale of change*
  - How particular actions reverberate and branch out into the industry
    - inaction, rumor, sentiment
  - We are going to try to observe, identify, predict causality, and ultimately – trade
- When considering **what** can change in a market it's important to understand the **economic foundations** that companies and retail investors focus on when trying to identify the correct risk/reward ratio
  - Understanding Risk takes two approaches
    - A wider approach of the entire environment
    - A more specific analysis of the individual security in the environment
- This course **must** focus on both approaches
  - Let's learn about them in a bit more detail

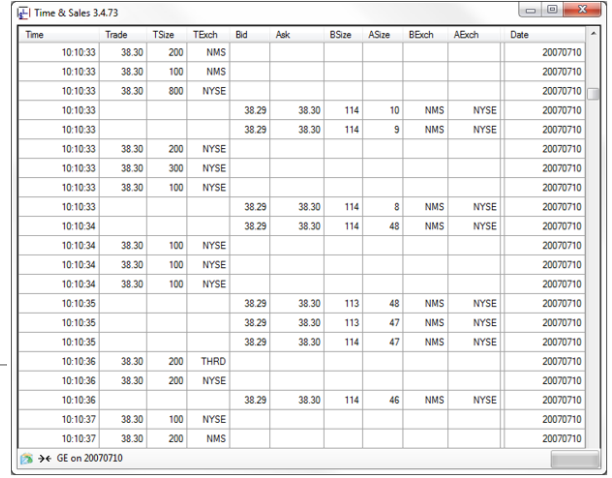
From Finance to Systems

From Finance to Systems

Moving from Static Balance sheets to dynamic data streams

Ticks as events

Tick Data



Time & Sales 3.4.73

Time	Trade	TSize	TExch	Bid	Ask	BSize	ASize	BExch	AExch	Date
10:10:33	38.30	200	NMS							20070710
10:10:33	38.30	100	NMS							20070710
10:10:33	38.30	800	NYSE							20070710
10:10:33				38.29	38.30	114	10	NMS	NYSE	20070710
10:10:33				38.29	38.30	114	9	NMS	NYSE	20070710
10:10:33	38.30	200	NYSE							20070710
10:10:33	38.30	300	NYSE							20070710
10:10:33	38.30	100	NYSE							20070710
10:10:33				38.29	38.30	114	8	NMS	NYSE	20070710
10:10:34				38.29	38.30	114	48	NMS	NYSE	20070710
10:10:34	38.30	100	NYSE							20070710
10:10:34	38.30	100	NYSE							20070710
10:10:34	38.30	100	NYSE							20070710
10:10:35				38.29	38.30	113	48	NMS	NYSE	20070710
10:10:35				38.29	38.30	113	47	NMS	NYSE	20070710
10:10:35				38.29	38.30	114	47	NMS	NYSE	20070710
10:10:36	38.30	200	THRD							20070710
10:10:36	38.30	200	NYSE							20070710
10:10:36				38.29	38.30	114	46	NMS	NYSE	20070710
10:10:37	38.30	100	NYSE							20070710
10:10:37	38.30	200	NMS							20070710

GE on 20070710

The following information is associated with a *trade*

- Stock or contract symbol, price, size(number of shares), Market Exchange(where the trade occurred) and a Timestamp

An individual trade is referred to as a **tick**

Tick data is a stream of trades that may occur while the market is open

- Ex: a series of trades for GE:

During any particular day, there may be thousands if not **millions** of trades for a particular financial symbol

Tick Data

There is a trade at 8:14.59 AM at the price of 1266.50. The following tick data represents a set of trades from 8:15.01 to 8:20 AM:

This means that 2 persons (one buyer and one seller) exchanged money for a stock, contract, option, etc.

At 8:15.00,		Open is 1266.50
At 8:15.23	7	units are traded for **1266.25**
At 8:16.23,	10	units are traded for 1266.75
At 8:17.47,	2	units are traded for 1267
At 8:18.11,	10	units are traded for **1267.50**
At 8:19.34,	3	units are traded for 1267.25
At 8:19.51,	10	units are traded for **1267.00**

Low: Lowest traded during a given period.

High: Highest traded during a given period.

Close: Last traded price during a given period.

The **Volume** refers to the total number units traded during this period. A unit may be a share of stock, a contract, an option, etc.

- During the 5 minute period, 7+10+2+10+3+10 or 42 units were traded.

So during the 8:15.01 to 8:20 AM period the opening price was 1266.50, the highest traded price was 1267.50, the lowest traded price was 1266.25, and the last traded price was 1267.00. There were 42 units traded during this period.

Open	High	Low	Close	Vol	Time	Date
1266.50	1267.25	1266.25	1267.00	42	8:20	20010604

The Anatomy of a Market Event

We look at prices on google for a particular stock and often, we read a lie
- We see that NVDA is trading at $150.00

The truth is, this is a gravestone
- It marks where it **last** traded
- This is no guarantee of future trade price

Who
- Buyer/seller

What
- Agreement on price, disagreement on value

Where
- Some exchange

When
- Microsecond timestamp

Why?
- Unknown, but we can infer: momentum? News? Fat finger? Algorithm?

AI Orchestration frameworks

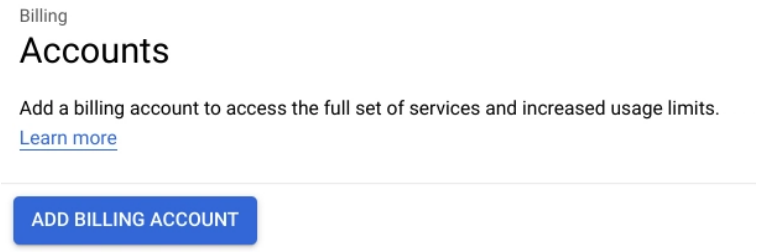
AI Orchestration Frameworks

- Imagine you are running a tournament
  - You need a system that keeps track of all of the players, the rules, the tools people use, etc
- An orchestration framework is like your tournament organizer
  - It makes sure every agent knows their job and communicates with others then things change
- Why does this matter?
  - Without orchestration
    - Agents might duplicate work
    - Miss important steps
    - Work against each other
- The framework connects agents to the right data, tools, and APIs
  - Helps agents remember that happened before
  - Learn from past mistakes

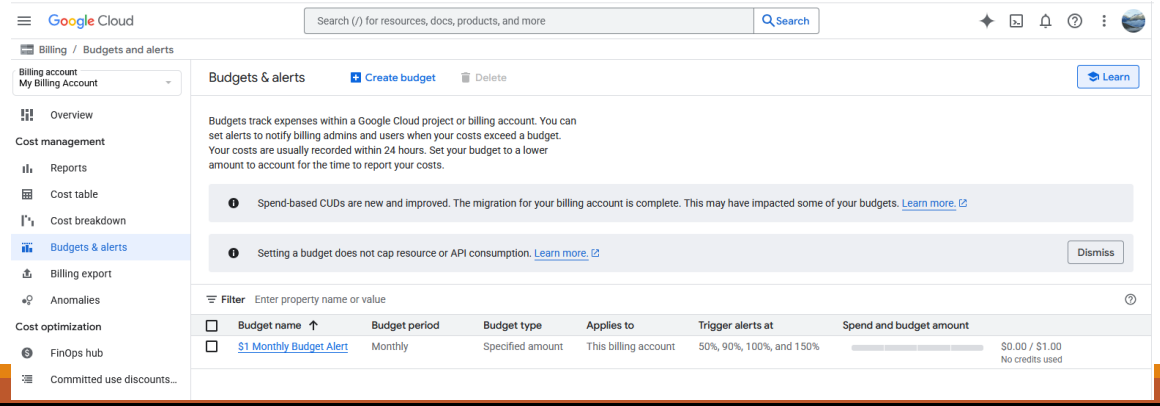
N8N

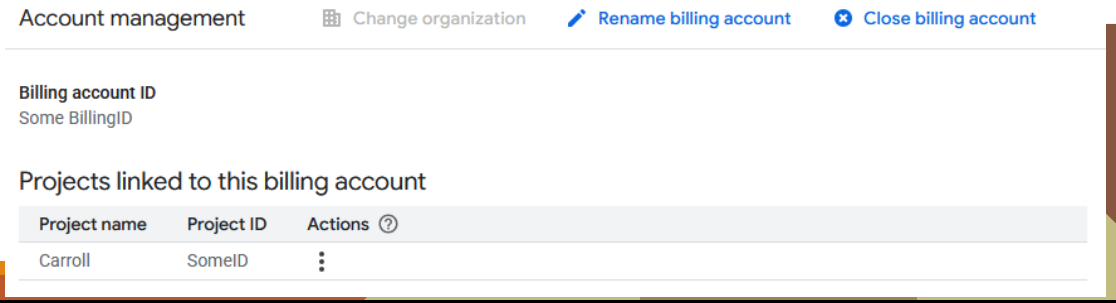
- N8N is an AI Orchestration framework that enables multiple different workflow creations
  - Why do I like it?
    - It’s free
      - We can locally host it
      - We can also use the education discount
  - Unlike other solutions, it allows us to build multiple agents together
- N8N allows us to build
  - Work flows
  - Chat flows
  - Chatbots
  - Agents
  - Text generators
- Let’s see what this looks like
  - Let’s setup our accounts

Setting up N8N: Part 1



- Setting up a Google Cloud Compute profile
  - We need to create a Virtual Machine that we can use to *host* our N8N server
- 1. Click here
- 2. Set up a billing account
  - The first step is to create a billing account. Google requires this in the case we go over the *free tier* threshold, but fear not, we will take steps to avoid charges.
  - To do this, there is a pane on the left labeled Billing, we simply click on **add billing account**
- Select your billing account which will navigate you to a budget overview pane, you can then select budgets and alerts, then create budget following the steps to set up a $1 budget alert



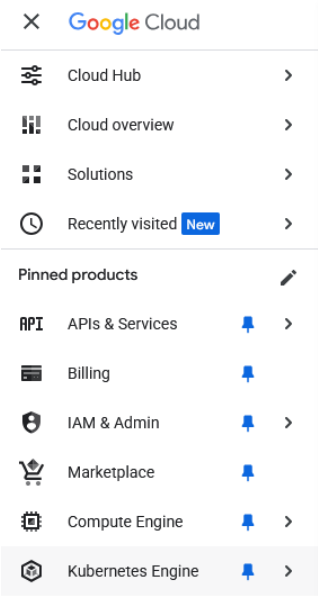


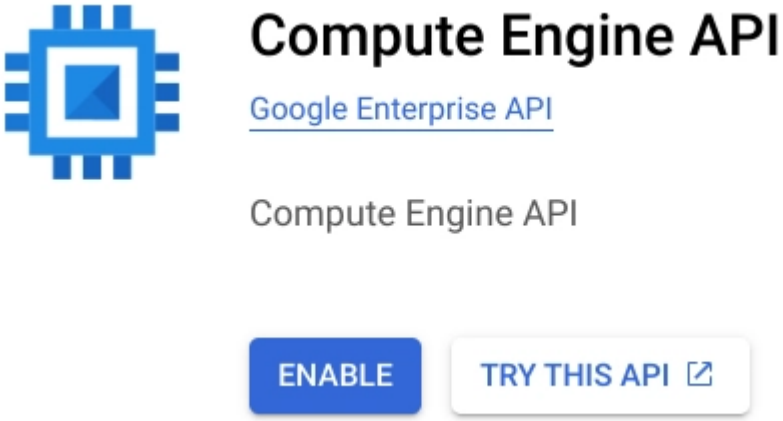
Setting up N8N part 2

Compute Engine	• 1 non-preemptible `e2-micro` VM instance per month in one of the following US regions:
	  • Oregon: `us-west1`
	  • Iowa: `us-central1`
	  • South Carolina: `us-east1`
	• 30 GB-months standard persistent disk
	• 1 GB of outbound data transfer from North America to all region destinations (excluding China and Australia) per month

- **Setting up a GC Virtual Machine**
- To remain in the free tier, google gives us exact specifications on what constitutes a free VM. **please** review the requirements before preceding.

- Select the navigation view located on the top left, select **Compute Engine**. Then simply click **enable**





Setting up N&N part 2

Region *
us-central1 (Iowa)
Region is permanent

Zone *
Any
Google will choose a zone on your behalf, maximizing VM obtainability. Zone is permanent.

New C4 machine types with Titanium SSD and Xeon 6 (Granite Rapids) Generally Available

Try new C4 machine types, including local SSD and bare metal variants, on the latest — Try now

General purpose | Compute optimized | Memory optimized | Storage optimized | GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series	Description	vCPUs	Memory	CPU Platform
○	C4	Consistently high performance	2 - 288	4 - 2,232 GB	Intel Emerald Ra
○	C4A	Arm-based consistently high performance	1 - 72	2 - 576 GB	Google Axion
○	C4D	Consistently high performance	2 - 384	3 - 3,072 GB	AMD Turin
○	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Ra
○	C3	Consistently high performance	4 - 192	8 - 1,536 GB	Intel Sapphire Ra
○	C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
◉	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Intel Broadwell
○	N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade La
○	N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD Milan
○	T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra
○	T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD Milan
○	N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Haswell

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. Learn more

Preset | Custom

e2-micro (2 vCPU, 1 core, 1 GB memory)

vCPU	Memory
0.25-2 vCPU (1 shared core)	1 GB

Networking

Firewall

Select firewall rules to allow specific network traffic from the Internet. Firewall rules will only be applied when an instance is created.

- [x] Allow HTTP traffic
- [x] Allow HTTPS traffic
- [ ] Allow Load Balancer Health Checks

Network tags

Network tags: https-server × http-server ×

Hostname

Hostname

Set a custom hostname for this instance or leave it default. Choice is permanent

Operating system
Ubuntu

Version *
Ubuntu 25.04 Minimal
x86/64, amd64 plucky minimal image built on 2025-07-08

Boot disk type *
Standard persistent disk

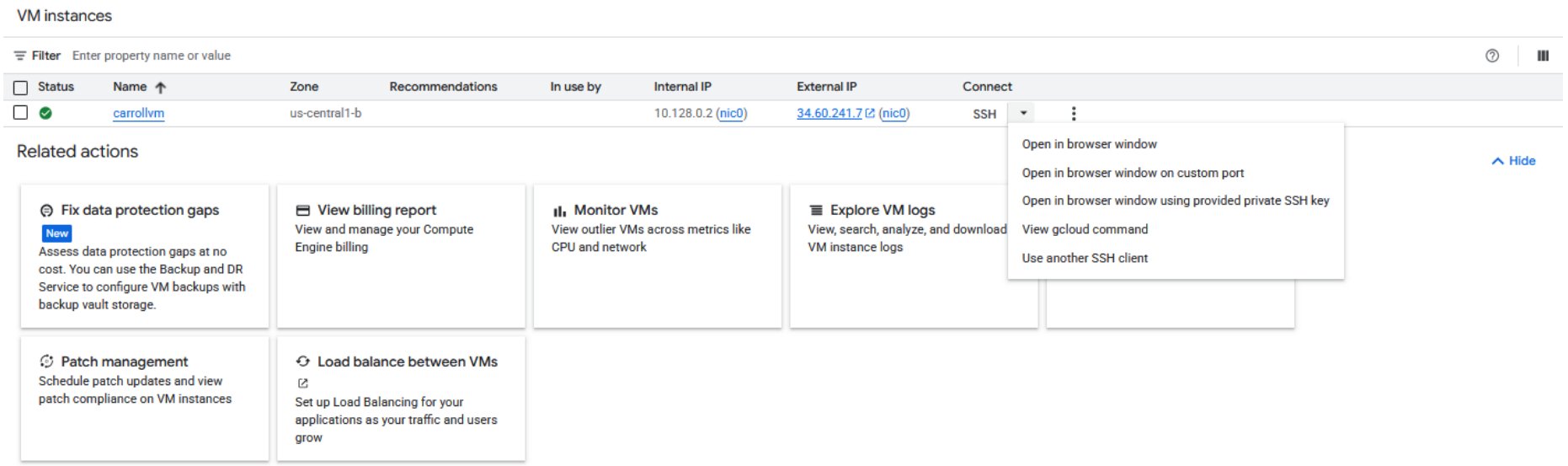
Compare disk types

Size (GB) *
30
Provision between 10 and 3072 GB

Show advanced configuration

Select | Cancel

Accessing the VM



Generating a DNS name for our VM

- DNS: **Domain Name System**

people: many identifiers:

- SSN, name, passport #

Internet hosts, routers:

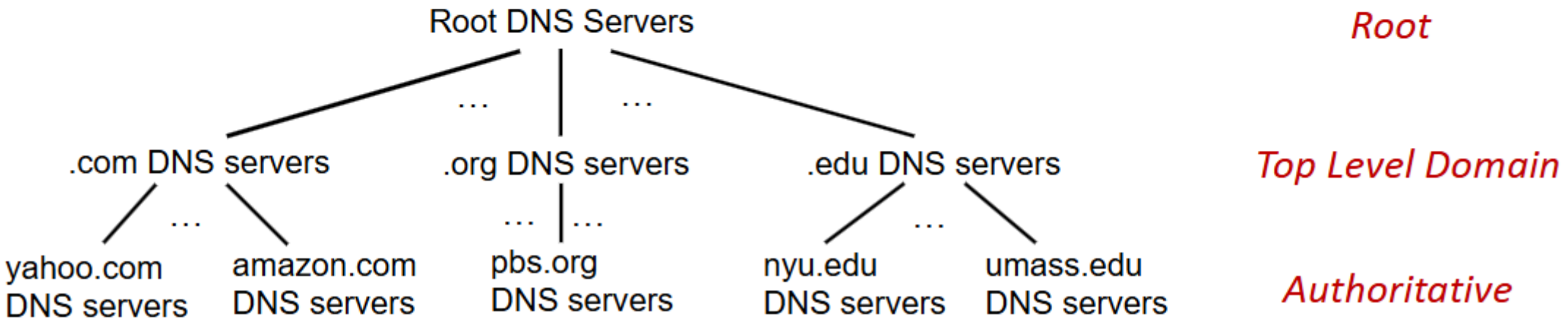
- IP address (32 bit) - used for addressing datagrams
- "name", e.g., cs.umass.edu - used by humans

Q: how to map between IP address and name, and vice versa ?

Domain Name System:

- *distributed database* implemented in hierarchy of many *name servers*
- *application-layer protocol:* hosts, name servers communicate to *resolve* names (address/name translation)
  - note: core Internet function, *implemented as application-layer protocol*
  - complexity at network's "edge"

DNS: a distributed, hierarchical database



Client wants IP address for www.amazon.com; 1st approximation:

- client queries root server to find .com DNS server
- client queries .com DNS server to get amazon.com DNS server
- client queries amazon.com DNS server to get IP address for www.amazon.com

Inserting a Record into a DNS

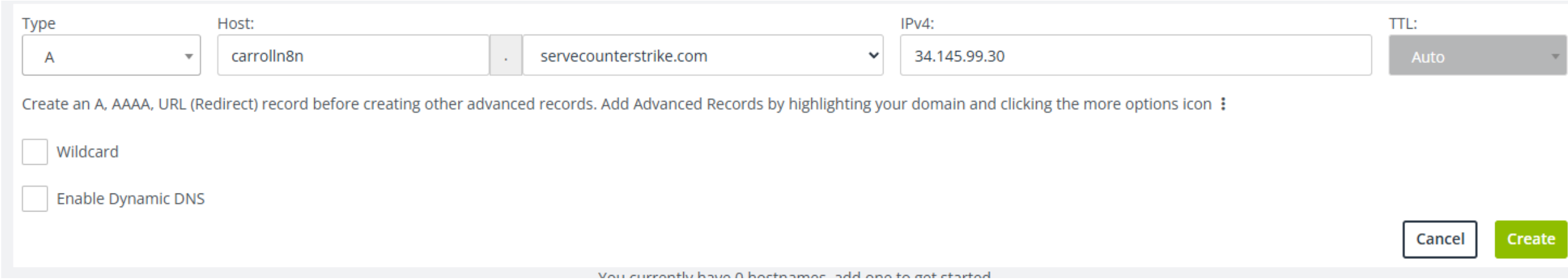
- Our goal is to register some domain name with a **DNS registrar**
- We can do this for free using
  - https://www.noip.com/
- The hostname does not matter much
  - Feel free to name it whatever you want!
- The IPv4 should be the IP address of your machine
  - How do we find this?

Finding our machines IP Address

- Two main ways
  - 1. We can use google's UI
    - BOOO BOOO
  - 2. We can use curl
    - Curl ifconfig.me
    - Let’s see how this is done!

```
jrfaster@n8ninstance:~$ curl ifconfig.me
34.145.99.30jrfaster@n8ninstance:~$
```

A valid DNS entry should look something like this



Installing N8N

- Full Guide found on canvas

n8n

- Overview
- Personal
- Chat beta

Overview

All the workflows, credentials and data tables you have access to

Create workflow

Prod. executions	Failed prod. executions	Failure rate	Time saved	Run time (avg.)
0	0	0%	--	0s

Workflows | Credentials | Executions | Variables | Data tables

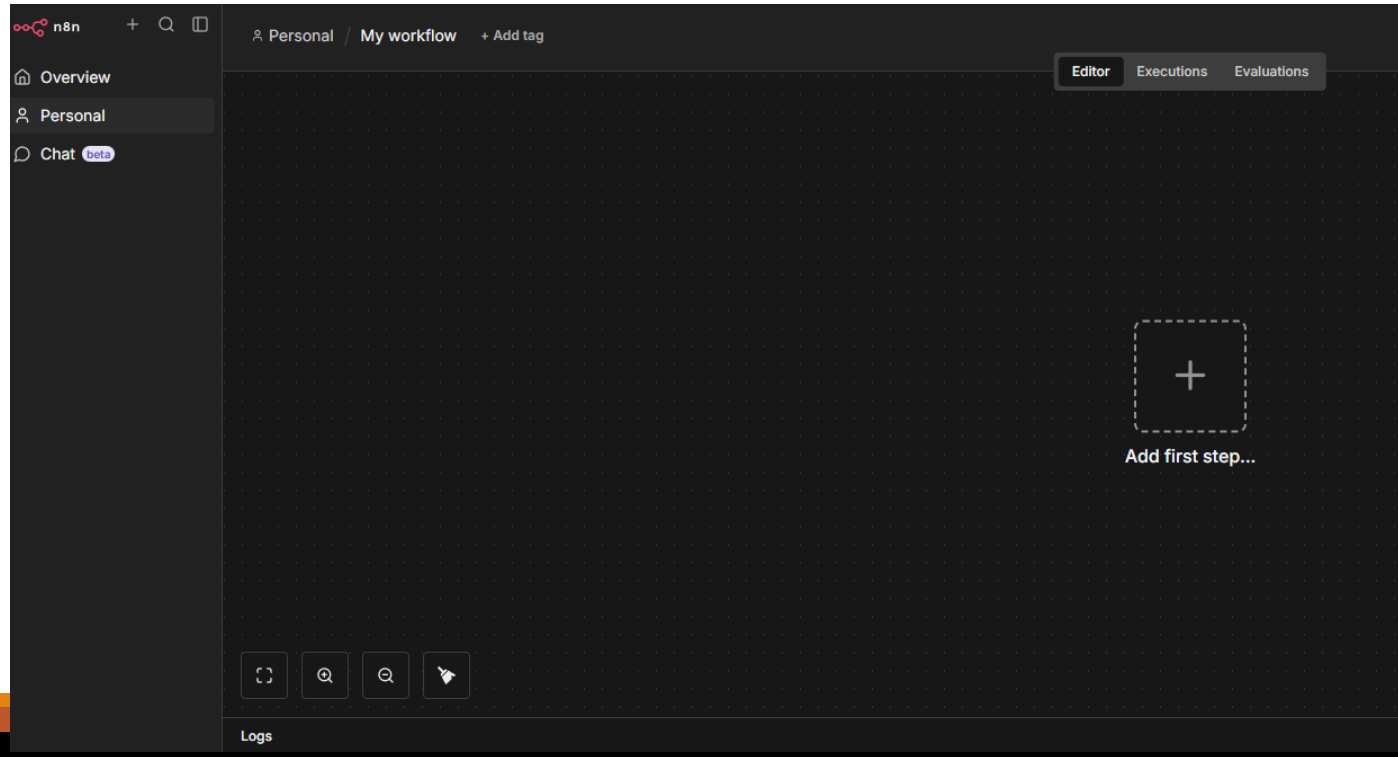
Search | Sort by last updated

Archived workflows are hidden in this view. Update filters

- Templates
- Insights
- Help
- Settings

Understanding the Canvas

- Select “Create Workflow”
- Let’s walk through this canvas starting with our left panel first



Left Side Panel

- On the left side of the **Editor UI**, there is a panel which contains the core functionalities and settings for managing your workflows. Expand and collapse it by selecting the small arrow icon.
  - **Overview**: Contains all the workflows, credentials, and executions you have access to. During this course, create new workflows here.
  - **Personal**: Every user gets a default personal project. If you don't create a custom project, your workflows and credentials are stored here.
  - **Projects**: Projects let you group workflows and credentials together. You can assign roles to users in a project to control what they can do. Projects aren't available on the Community edition.
  - **Admin Panel**: n8n Cloud only. Access your n8n instance usage, billing, and version settings.
  - **Templates**: A collection of pre-made workflows. Great place to get started with common use cases.
  - **Variables**: Used to store and access fixed data across your workflows. This feature is available on the Pro and Enterprise Plans.
  - **Insights**: Provides analytics and insights about your workflows.
  - **Help**: Contains resources around n8n product and community.
  - **What's New**: Shows the latest product updates and features.

Top Bar

- **Workflow Name**: By default, n8n names a new workflow as "My workflow", but you can edit the name at any time.
- **+ Add Tag**: Tags help you organize your workflows by category, use case, or whatever is relevant for you. Tags are optional.
- **Publish**: This button publishes the current workflow. By default, workflows are not published.
- **Share**: You can share and collaborate with others on workflows on the Starter, Pro, and Enterprise plans.
- **Save**: This button saves the current workflow.
- **History**: Once you save your workflow, you can view previous versions here.

Canvas

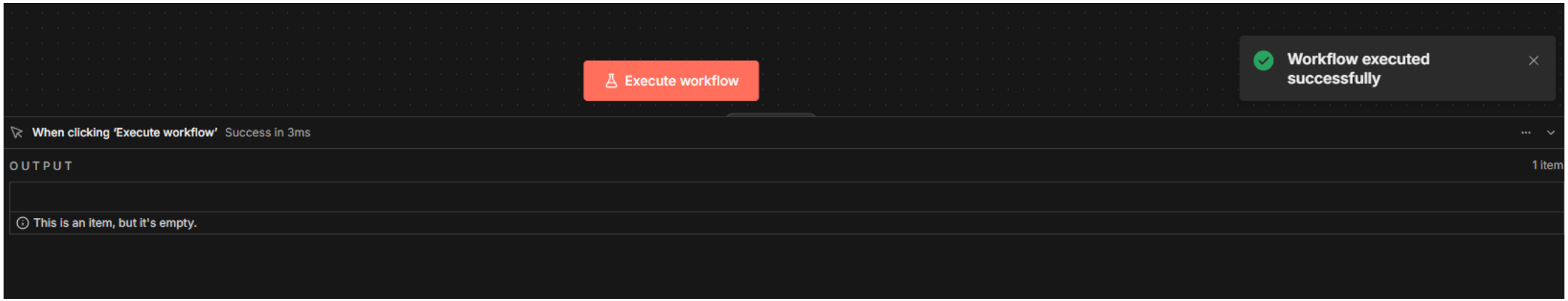
- The **canvas** is the gray dotted grid background in the Editor UI. It displays several icons and a node with different functionalities:
  - Buttons to zoom the canvas to fit the screen, zoom in or out of the canvas, reset zoom, and tidy up the nodes on screen.
  - A button to **Execute workflow** once you add your first node. When you click on it, n8n executes all nodes on the canvas in sequence.
  - A button with a **+** sign inside. This button opens the nodes panel.
  - A button with a note icon inside. This button adds a sticky note to the canvas (visible when hovering on the top right + icon).
  - A dotted square with the text "Add first step." This is where you add your first node.

Nodes

- In graph theory, a node represents a specific point in a network where data is processed.
  - You can think of a node as a station on an assembly line. At each station, a worker (or machine) performs a specific task, pausing to inspect, modifying the product, or passing it quickly to the next stage.
- In n8n, nodes are the **functional units** that give our system power.
  - It is through the strategic arrangement of these nodes that we transform our ideas into agentic systems.
- Based on their function, n8n classifies nodes into four types:
  - **App** or **Action Nodes**
  - **Trigger Nodes**
  - **Core Nodes**
  - **Cluster Nodes**
- You can open the node panel:
  - Click the **+** icon in the top right corner of the canvas.
  - Click the Tab key on your keyboard.

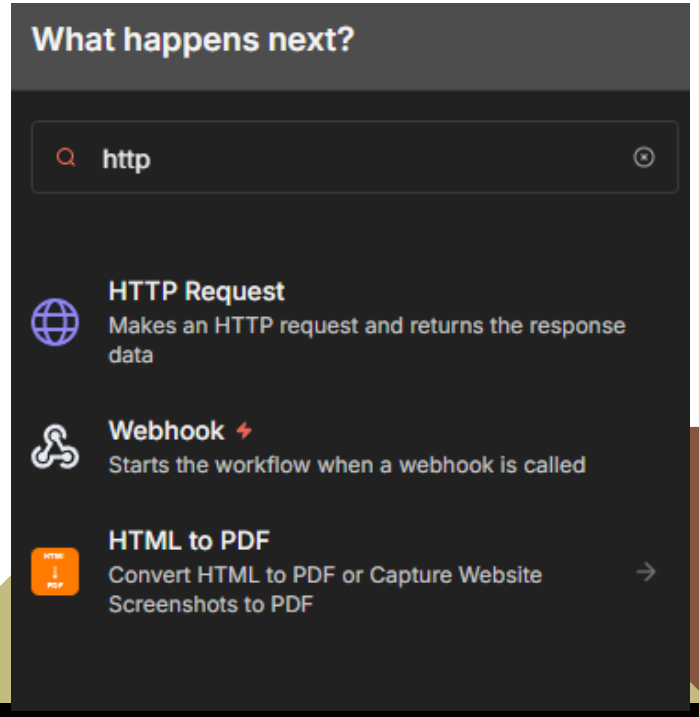
Executing a node or workflow

- Node buttons
- If you hover on a node, you'll notice that three icons appear on top:
  - Execute the node (Play icon)
  - Deactivate/Activate the node (Power icon)
  - Delete the node (Trash icon)


Execute workflow
Workflow executed successfully
When clicking 'Execute workflow' Success in 3ms
OUTPUT
1 item
This is an item, but it's empty.

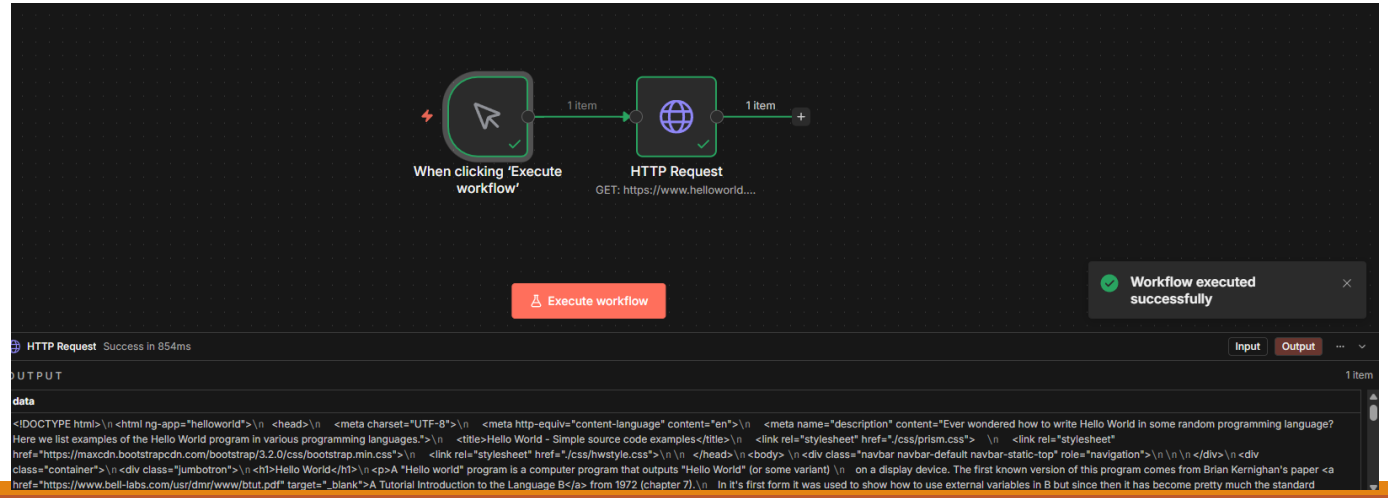
HTTP Node Request

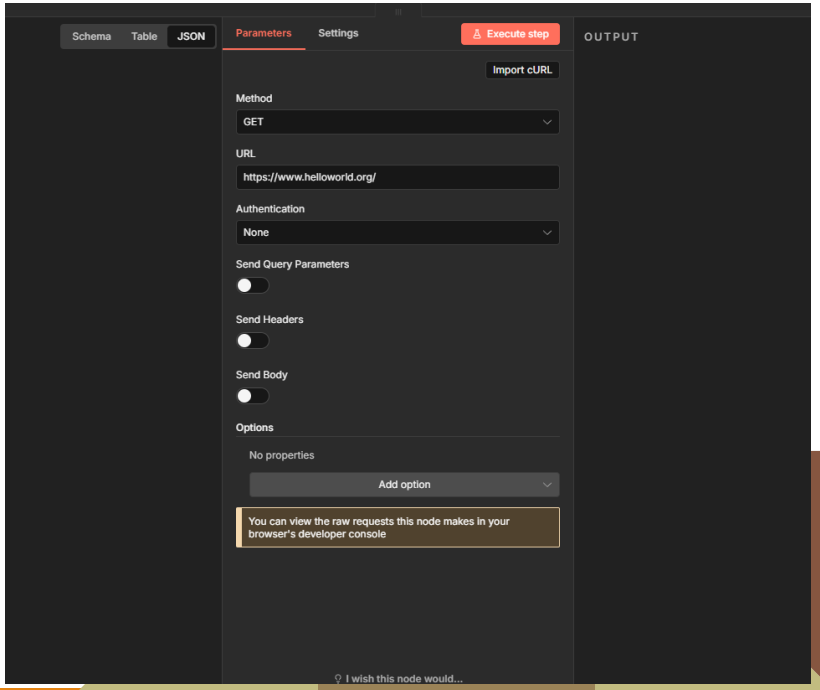
- Press “tab” or click the plus icon to open the node panel
  - Search for HTTP Request
- **HTTP Requests**: Fundamental web protocol for asking servers for data.
  - When you type a URL into your browser, you are essentially sending a request to a server asking for information.



Get Requests

- **GET Request**: Retrieves data without changing the server; most common method.
- **N8N HTTP Node**: Acts like a browser to fetch data for workflows.
- **HTML Output**: Responses come as raw HTML;





Just like that, our first workflow!

- Huzzah, we have significantly complicated what was a simple process
  - Welcome to computer science!
    - 15 hours to automate something you could have done manually in 10 minutes

So now what?

- N8N is our *orchestration framework*
  - It's main goal is to simply **automate** processes
  - Modern field of **Automation Experts**
- Generally
  - We want to use N8N, or similar systems, to automate repeatable tasks
- Throughout this semester, we are going to use N8N and other tools to try and automate decision and trading principles with a specific focus on the US stock market(maybe crypto, maybe prediction markets)
- **Warning**
  - We are going to fail
  - N8N is a **new** technology
    - Much of what we are trying to do is experimental.
      - **I want you to experiment. There is no correct answer**
      - **Learn, Learn, Learn, Learn, Learn – Try whatever**

Our First Experiment

MOVING AVERAGES

How do investors make buy and sell decisions?

This is perhaps the **golden** question of what makes a *trader*

There are many ways that people make buying and selling decisions

- What to buy(or sell)
- When to buy(or sell)

It is possible, and more common, to weave these two decisions together into a single model as you analyze buy and sell signals

- But first, let's consider how we might even **get** the price of a particular security

Alpaca

PAPER TRADING

Alpaca

Alpaca allows us to execute **paper trades**

Create an account:

- https://app.alpaca.markets/brokerage/new-account

What is an API?

- API: Application Protocol Interface
  - This is a set of rules and protocols that allows different software programs to communicate with each other and exchange data
  - It's the translator that describes how some software X can request specific data from software Y

Ultimately, it is an abstraction, just like N8N is

- It abstracts the inner mechanisms
  - How the data is maintained
  - How the data is ultimately returned
  - How the host machine operated
    - OS, language, etc.

Ask/Response format

- If you pose a properly formatted API request, the expectation is you get a response
  - Code 200
- If you pose an incorrect API request, you often get an error code, Each with their own meanings
  - Code 4xx – client side error(you)
  - Code 5xx – server error(them)

Using Alpaca’s API to Grab Data

API Reference

- https://docs.alpaca.markets/reference/issuetokens

Alpaca allows us to grab historic data over a given set of dates for the provided securities

Let’s Structure our goals

1. Get a given security from a user
2. Identify the correct API route to ping
3. Form our *ask* request

Price of the security

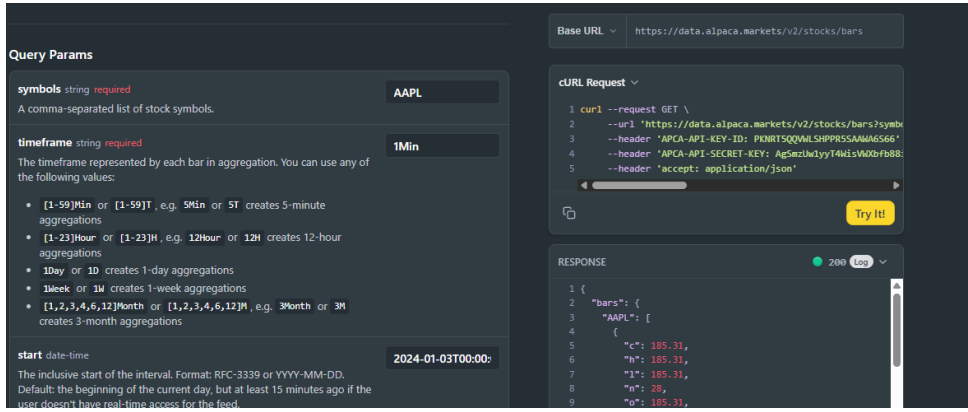
Using Alpaca, we can grab the price, take some time to try and find the correct api endpoint

Alpaca allows us to test a given API which makes placing it into N8N seamless

The **C** is the security price at the end of the tick

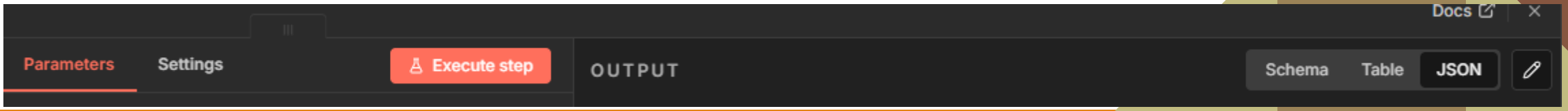
- More on the rest next time

We can *copy* the curl request and *import* it directly into n8n



Configuring Nodes

- Each node *may* have it's own unique fields that require editing
  - Some common *themes* persist between nodes
- When you add a new node to the Editor UI, the node is automatically activated. The node details will open in a window with separate options
  - **Parameters**
    - This is information that we want to pass **in** to our node
  - **Settings**
    - These typically allow us to control the nodes design and execution flow
  - **Docs**
    - Should we need, we can go to the official n8n documentation to understand more about the node



Parameters

- A **Parameter** is always going to be unique to the specific node and *environment* we are working in
  - Our manual input trigger node does not require a parameter
  - The node docs will often help you understand what the requirements might be
- When we import a curl request, these fields are auto filled
  - They can be manually changed later
    - We will get to this in a bit